

Sales Data Exploratory Data Analysis (EDA)

Project Overview

This project was completed as part of my **Data Analytics Internship at Decode Labs**.

The objective of this project was to perform **Exploratory Data Analysis (EDA)** on a sales dataset to understand patterns, trends, product performance, descriptive statistics, and potential outliers.

The project focuses on transforming cleaned sales data into meaningful analytical insights through statistical analysis and interactive visualizations.

Project Objectives

The main objectives of this project were to:

- Calculate basic descriptive statistics such as **Mean, Median, and Count**
 - Analyze sales and revenue trends
 - Compare revenue performance across products
 - Analyze transaction/record counts by product
 - Identify potential outliers using **Z-score**
 - Identify important patterns and observations from the dataset
 - Present the findings through an interactive dashboard
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Tools & Technologies

- **Microsoft Excel** — Data cleaning, preparation, statistical calculations, and outlier analysis
 - **Microsoft Power BI** — Interactive dashboard and data visualization
 - **GitHub** — Project documentation and submission
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Analysis Performed

1. Descriptive Statistics

The following statistics were calculated:

- Mean
- Median
- Count

The comparison between the mean and median was used to understand the distribution of Total Price values and whether higher-value observations may be influencing the average.

2. Trend Analysis

A trend visualization was created to analyze changes in the selected sales metric over time.

The dashboard includes year-based filtering for:

- 2023
- 2024
- 2025

3. Product Analysis

Product-level analysis was performed to compare:

- Revenue by product
- Transaction/record count by product

This helped identify products with relatively higher and lower sales performance.

4. Outlier Analysis

Potential product-level outliers were identified using the **Z-score method**.

A value was treated as a potential outlier when:

Z-score > +3 or Z-score < -3

The outliers were reviewed as unusual observations rather than automatically being removed.



Dashboard

The Power BI dashboard contains:

- Total Quantity
- Total Revenue
- Mean Total Price
- Median Total Price
- Record Count
- Year filters
- Order Status filter
- Trend Analysis
- Revenue by Product
- Transaction Count by Product

- Product-Level Outlier Analysis

The dashboard also includes a separate **Insights** page connected through page navigation.

Key Insights

Some of the key observations from the analysis include:

- The **mean Total Price is higher than the median**, suggesting that higher-value transactions may be pulling the average upward.
 - Product-level analysis shows differences in both revenue and transaction volume across products.
 - Potential outliers were identified using the ± 3 Z-score threshold and should be investigated further rather than automatically treated as incorrect data.
 - Revenue shows fluctuations over time, with periods of higher and lower performance.
 - Product performance varies across the selected years.
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Project Files

The repository contains the following project materials:

- `README.md` — Project documentation
 - `Sales_Data_Cleaned.xlsx` — Cleaned dataset and analysis
 - `Sales_EDA.pbix` — Power BI dashboard
 - `Screenshots/` — Dashboard screenshots
 - `Documentation/` — Additional project documentation and insightsPROJ
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Methodology

The project followed this general workflow:

Raw Data → Data Cleaning → Statistical Analysis → Outlier Detection → Data Visualization → EDA Insights → Interactive Dashboard

Learning Outcomes

Through this project, I practiced:

- Data cleaning and preparation
- Descriptive statistics
- Exploratory Data Analysis
- Z-score based outlier detection

- Trend analysis
 - Product-level analysis
 - Power BI dashboard development
 - Data storytelling and insight generation
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